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## **Developing a Robust Intelligent Model for Predicting Minimum Miscibility Pressure in Hydrocarbon Gas - Oil Systems: Implication for Miscible Gas Enhanced Oil Recovery Process**

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### **Abstract**

Miscible gas-based enhanced oil recovery (EOR) methods are widely regarded as effective techniques for improving oil recovery when natural reservoir drive mechanisms become insufficient. However, the successful application of these methods relies on accurate prediction of key parameters, particularly the minimum miscibility pressure (MMP) in hydrocarbon gas–oil systems. The experimental determination of MMP is costly, time-consuming, and resource-intensive, making the need for reliable and rapid predictive tools more pressing. This study introduces two advanced machine learning (ML) models, namely artificial neural network (ANN) and extra trees (ET), to predict MMP in hydrocarbon gas–oil systems. A comprehensive experimental dataset consisting of over 100 datapoints of MMP, was utilized to train and validate the models. Various evaluation techniques, including statistical metrics, graphical analysis, trend evaluations, and the Shapley Additive exPlanations (SHAP) method, were employed to assess the robustness, accuracy, and interpretability of the models. The results demonstrated that the ANN model outperformed the ET model, achieving a root mean square error (RMSE) of only 1.1883 MPa. Graphical error analysis and trend comparisons further confirmed that the ANN model closely aligned with experimental MMP values across a range of operational conditions. Additionally, SHAP analysis highlighted the transparency and interpretability of the ANN model, providing insights into the influence of input variables on MMP predictions. The findings underline the advantages of using ML-based models for MMP prediction, offering significant improvements in terms of efficiency, cost-effectiveness, and operational optimization. These models can aid in making appropriate decisions about the timing and volume of hydrocarbon gas injection in EOR processes, contributing to more effective planning and execution of gas injection strategies.

**Keywords:** Hydrocarbon gas, Enhanced oil recovery, Minimum miscibility pressure, Miscibility, Machine learning, Artificial neural network

## Introduction

The world's energy landscape remains heavily reliant on hydrocarbon resources. Oil, coal, and natural gas continue to serve as the pillar of modern civilization, powering transportation, manufacturing, electricity generation, and heating. According to recent statistics, oil alone accounts for approximately 36% of the global energy supply, followed by coal at 27.4% and natural gas at 23% [1]. This dependency persists despite growing efforts to diversify energy portfolios. As global population growth, rapid urban development, and industrial expansion continue to fuel energy demand [2,3], the efficient utilization of existing hydrocarbon reserves becomes ever more critical. In response, the petroleum industry has increasingly turned to Enhanced Oil Recovery (EOR) technologies to maximize output from mature or depleted reservoirs [4–6].

Among the various EOR techniques available, such as thermal recovery, chemical injection, and water flooding, gas injection techniques have emerged as particularly appealing due to their operational efficiency and potential for high recovery rates [7–10]. In both miscible and immiscible forms, gas injection uses agents such as associated gas, nitrogen, flue gas, and carbon dioxide (CO<sub>2</sub>), to displace trapped oil and improve reservoir sweep efficiency. However, for miscible gas injection to be successful, a crucial pressure threshold known as the Minimum Miscibility Pressure (MMP) must be achieved. MMP is defined as the lowest pressure at which the injected gas becomes fully miscible with reservoir oil, resulting in a single-phase fluid that enhances oil displacement and flow [11–15].

Traditionally, determining MMP has required sophisticated experimental setups such as the slim tube test, vanishing interfacial tension (VIT) measurements, or the rising bubble apparatus (RBA) [4,16]. While these methods are known for their accuracy, they are also resource-intensive, time-consuming, and operationally challenging, particularly in high-pressure, high-temperature conditions or when multiple gas compositions must be evaluated. To circumvent these limitations, researchers have proposed various empirical and semi-empirical correlations to estimate MMP. However, despite their simplicity and computational efficiency, these correlations often fall short in terms of accuracy and adaptability across different fluid systems and gas compositions [6].

In recent years, advances in artificial intelligence (AI) and machine learning (ML) have opened new avenues for predictive modeling in the oil and gas sector [17,18]. These techniques are especially well-suited for problems characterized by complex, nonlinear relationships and comprehensive datasets, both of which apply to MMP prediction. Several studies have leveraged ML models such as artificial neural networks (ANNs), hybrid intelligent systems, and evolutionary optimization algorithms to improve the estimation of MMP [4].

The present study investigates and compares the performance of two powerful yet distinct machine learning algorithms, namely Extra Trees (ET) and Artificial Neural Networks (ANN), for MMP prediction in CO<sub>2</sub> – oil systems. Beyond simply benchmarking their predictive accuracy, this work incorporates explainable AI (XAI) techniques to assess the interpretability of the models. Specifically, the SHAP methodology is applied to evaluate feature contributions and ensure the predictions align with established physical trends. Through this integrated approach, we aim to deliver a predictive framework that is not only accurate and robust, but also transparent and physically meaningful for field-scale implementation in EOR design.

## Methodology

In the present investigation, a rigorous predictive framework was established to estimate the Minimum Miscibility Pressure (MMP) in hydrocarbon gas-oil systems using two robust machine learning (ML) algorithms: Artificial Neural Network (ANN) and Extremely Randomized Trees (ET). These models were not chosen arbitrarily but selected due to their proven capacity to address high-dimensional and nonlinear data challenges, as demonstrated in numerous prior studies [18,19]. ANN, inspired by the architecture of the human brain, is known for its adaptive learning capabilities and ability to generalize complex relationships

even when the underlying physical correlations are not explicitly understood. ET, on the other hand, represents an ensemble learning strategy that enhances prediction robustness by aggregating uncorrelated decision trees, offering an effective remedy against overfitting and data variability.

Before developing these ML models, special attention was given to curating a high-quality and diverse experimental dataset, which serves as the foundation for any meaningful data-driven prediction. The dataset used in this study comprises 118 experimentally determined MMP values, carefully extracted from multiple reliable sources spanning the published literature [20–30]. These measurements reflect a wide range of gas injection compositions, wherein methane (C1) and intermediate hydrocarbons (C2–C6) often constitute the predominant fractions, sometimes exceeding 50 mol%. Such broad compositional coverage ensures that the data reflect a rich spectrum of practical field scenarios, thereby increasing the reliability and generalizability of the developed models.

To describe the underlying data structure and provide a preliminary insight into the nature of the dataset, a comprehensive exploratory data analysis was performed. Table 1 summarizes the key statistical descriptors, such as minimum, maximum, mean, and standard deviation, for both input variables and the target output (MMP). To further enhance interpretability, a pairwise correlation plot (Fig. 1) was constructed, which not only visualizes the distribution of each parameter but also highlights interdependencies between input features and their potential influence on MMP (based on the red color lines). This step is critical in guiding model design and helps in understanding the degree of nonlinearity and variability present in the data.

Once the data were thoroughly validated and pre-processed, the implementation of the machine learning models began. A random data splitting approach was employed to partition the dataset into two subsets: 80% of the data points were allocated for training the models, while the remaining 20% served as a blend test set to evaluate generalization performance. This methodology ensures that the models are trained on a sufficiently large sample to learn complex relationships and are also tested on previously unseen data to objectively assess prediction accuracy. Importantly, multiple reruns were performed to reduce bias associated with a single split and to obtain performance metrics that are more statistically robust.

**Table 1—Statistical description of the collected database**

|      | $x_{Vol}$ (%mol) | $x_{Int}$ (%mol) | $M_w$ C5+ (g/mol) | $y_{CO_2}$ (%mol) | $y_{N_2}$ (%mol) | $y_{C1}$ (%mol) | $y_{C2-C6}$ (%mol) | T (K)  | MMP (MPa) |
|------|------------------|------------------|-------------------|-------------------|------------------|-----------------|--------------------|--------|-----------|
| mean | 29.485           | 26.584           | 197.5695          | 5.329             | 2.881            | 50.152          | 41.592             | 374.29 | 23.313    |
| std  | 16.116           | 9.819            | 35.9381           | 10.500            | 11.031           | 32.373          | 31.544             | 19.86  | 9.741     |
| min  | 0.000            | 1.410            | 144.2290          | 0.000             | 0.000            | 0.000           | 0.000              | 312.59 | 2.758     |
| 25%  | 23.100           | 20.590           | 179.0385          | 0.000             | 0.000            | 31.500          | 20.000             | 357.59 | 16.996    |
| 50%  | 30.090           | 24.210           | 187.9166          | 1.050             | 0.225            | 50.000          | 40.000             | 383.15 | 24.752    |
| 75%  | 36.740           | 29.970           | 204.1228          | 6.390             | 0.360            | 73.050          | 55.458             | 388.71 | 30.144    |
| max  | 65.610           | 58.150           | 261.6562          | 50.000            | 50.000           | 100.000         | 100.000            | 444.26 | 47.229    |

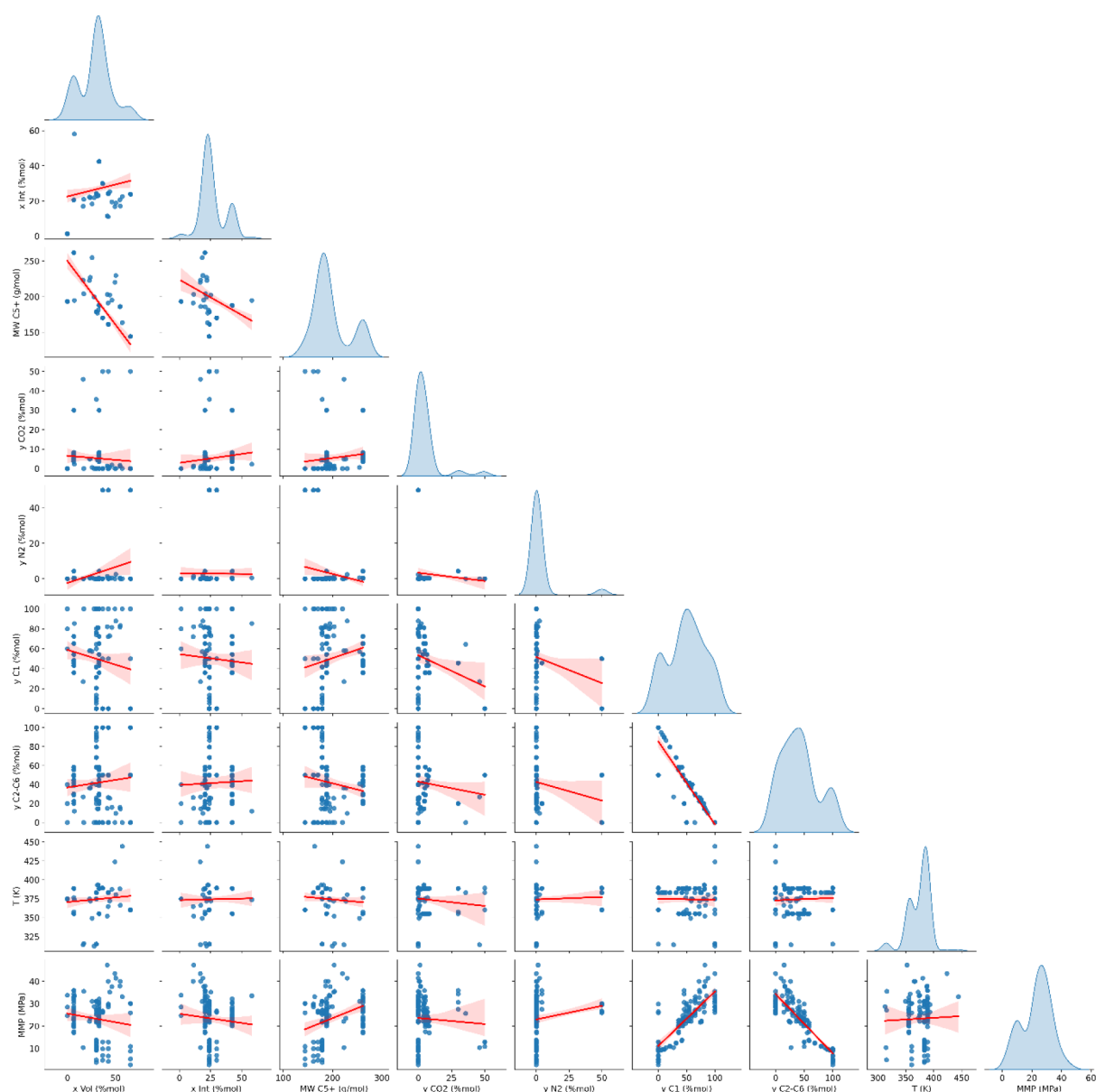


Figure 1—Pair plot showing the distribution of the data points

During the model development phase, extensive hyperparameter optimization was conducted to fine-tune each algorithm. For the ANN, two hidden layers involving 11 and 10 neurons, respectively, were identified through a trial-and-error process as the appropriate control parameters. In parallel, the suitable control parameters for the ET model were `n_estimators`: 373, `max_depth`: 19, `min_samples_split`: 3, and `min_samples_leaf`: 2. Both models were trained using loss minimization techniques appropriate for regression tasks, such as the mean squared error (MSE) loss function. It is worth mentioning that the aforementioned parameters have a great impact on the reliability of the built models and for this reason their appropriate selection was done thoroughly.

What distinguishes this work from existing literature is not only the methodological rigor but also the emphasis on creating models that are both accurate and interpretable. Rather than focusing solely on predictive performance, we also explored the relative importance of input variables and assessed how different compositions and thermodynamic parameters affect MMP. This approach ensures that the developed models are not "black boxes" but can offer valuable physical insights into the gas-oil miscibility

phenomenon, thereby assisting reservoir engineers in designing more effective enhanced oil recovery methods.

## Results and discussion

To rigorously evaluate the predictive capabilities of the constructed machine learning models, namely the ANN and ET, we adopted a comprehensive assessment strategy that blends both numerical metrics and visual diagnostics. This approach enables a thorough validation of the models' strengths in estimating MMP for hydrocarbon gas–oil systems under various thermodynamic and compositional conditions. The core of our accuracy evaluation begins with two widely accepted performance indicators in the machine learning domain: the Root Mean Square Error (RMSE) and the coefficient of determination ( $R^2$ ). These metrics, formally defined by Equations (1) and (2), offer insights into both the magnitude of error and the proportion of variance in the actual data that the model can account for.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (MMP_{i,exp} - MMP_{i,pred})^2} \quad (1)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (MMP_{i,exp} - \overline{MMP}_{exp})^2}{\sum_{i=1}^N (MMP_{i,exp} - \overline{MMP}_{exp})^2} \quad (2)$$

where  $MMP_{i,exp}$ ,  $MMP_{i,pred}$ ,  $\overline{MMP}_{exp}$ , and  $N$  correspond to the real minimum miscibility values, the MMP values estimated by the implemented paradigms, the mean of the measured MMP, and the total number of measurements, respectively. The accuracy of a model is deemed promising if its coefficient of determination approaches 1 and the RMSE tends toward zero.

Table 2 summarizes the RMSE and  $R^2$  values for both the training and testing phases of each model. Notably, both models exhibit commendable performance across the datasets. However, the ANN model clearly distinguishes itself by reaching an  $R^2$  value of 0.9934 and an impressively low RMSE of 1.1883 on the full dataset. These values signify that the ANN can capture nearly all the variance in MMP behavior while maintaining minimal predictive error. Although ET also achieved respectable performance, its metrics trailed those of the ANN model, suggesting that while ensemble tree-based approaches are effective, the ANN's ability to model complex nonlinearities yields superior precision in this case.

**Table 2—Statistical evaluation of the implemented ANN and ET models**

| Stage         | Metrics | ANN    | ET     |
|---------------|---------|--------|--------|
| Training data | $R^2$   | 0.9955 | 0.9431 |
|               | RMSE    | 0.9934 | 2.4973 |
| Test data     | $R^2$   | 0.9857 | 0.9348 |
|               | RMSE    | 1.7541 | 1.7978 |
| All data      | $R^2$   | 0.9934 | 0.9414 |
|               | RMSE    | 1.1883 | 2.3574 |

Beyond raw statistics, visual tools provide intuitive and immediate feedback about model behavior. One such method is the cross-plot, where predicted values are plotted against the experimentally obtained MMP data. Ideally, a model that performs perfectly would have its data points closely clustered around the 45-degree line (i.e., the identity line where predicted equals actual). Fig. 2 displays the cross-plots for both ANN and ET models. An outstanding observation from these plots is the higher density and tighter clustering of ANN data points along the identity line, compared to the somewhat more scattered alignment observed for ET. This visual evidence complements the statistical metrics, reinforcing that ANN offers a more reliable

estimation of MMP across the entire dataset. Moreover, the plot suggests that ANN remains robust in extrapolating predictions for both low and high MMP values, a critical feature for its application in field operations.

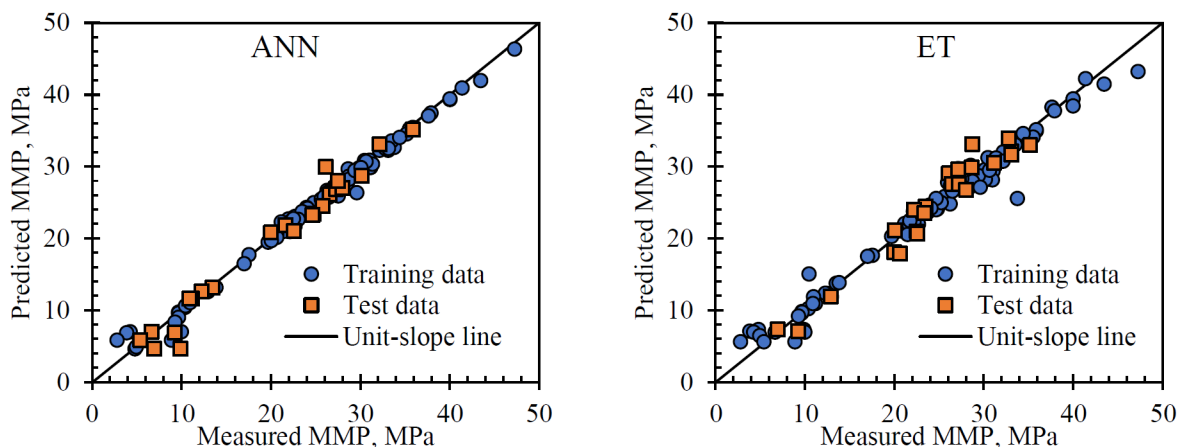


Figure 2—Cross plots of the established ANN and ET models

To further inspect model accuracy, Fig. 3 presents the residual distribution of predicted MMP values. Residuals, defined as the difference between predicted and actual values, help detect systematic errors or biases in model predictions. A well-performing model should yield residuals symmetrically distributed around zero, with minimal deviation. Both models exhibit relatively balanced residuals, though the ANN model's error distribution is noticeably tighter and more centered. This suggests that ANN not only reduces the magnitude of error but also avoids consistent underestimation or overestimation. Such behavior is crucial for real-world applications, where large residuals can translate to operational inefficiencies or misjudged design parameters.

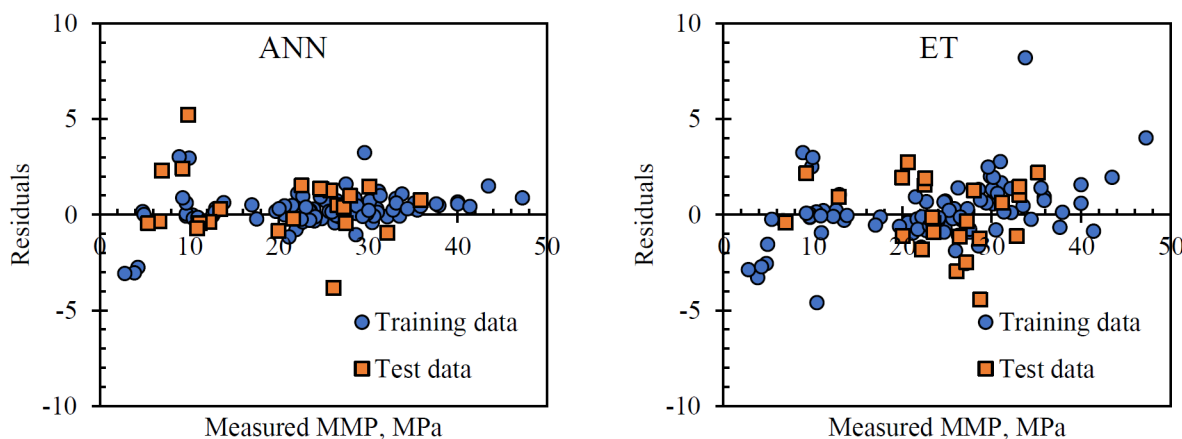


Figure 3—Residual plots of the established ANN and ET models

While achieving high predictive performance is vital, modern machine learning modeling must also address interpretability, especially in high-complex domains like the oil and gas industry. To this end, we employed SHapley Additive exPlanations (SHAP), a powerful technique rooted in game theory that attributes the contribution of each feature to the model's output. SHAP provides promising explanations, making it ideal for understanding complex models like ANN.

Fig. 4 illustrates the SHAP summary plot for the ANN model. Each point on this plot represents a SHAP value for a specific feature on a specific sample, with red indicating high feature values and blue indicating



low. Features are arranged from top to bottom in order of their overall impact on the model. A key finding from this analysis is the dominant role of the mole fractions of C2–C6 and C1 in the injected gas mixture. C2–C6, often associated with intermediate hydrocarbons, displayed a strong inverse relationship with MMP: as their mole fraction increases, the model tends to predict lower MMP values. This is consistent with the physical understanding that heavier hydrocarbons promote miscibility due to their higher solubility in the oil phase. Conversely, the presence of C1 (methane), the lightest hydrocarbon component, exerts an opposite effect. Higher concentrations of C1 are associated with increased MMP. This behavior is also visible in the SHAP summary: high C1 values (red points) are predominantly situated on the right-hand side, indicating a positive contribution to MMP. SHAP analysis thus provides not only confirmation of known physical phenomena but also a quantitative aspect to assess feature contributions across the entire dataset. This enhances confidence in the model's decisions and opens the door for explainable, transparent deployment in field operations.

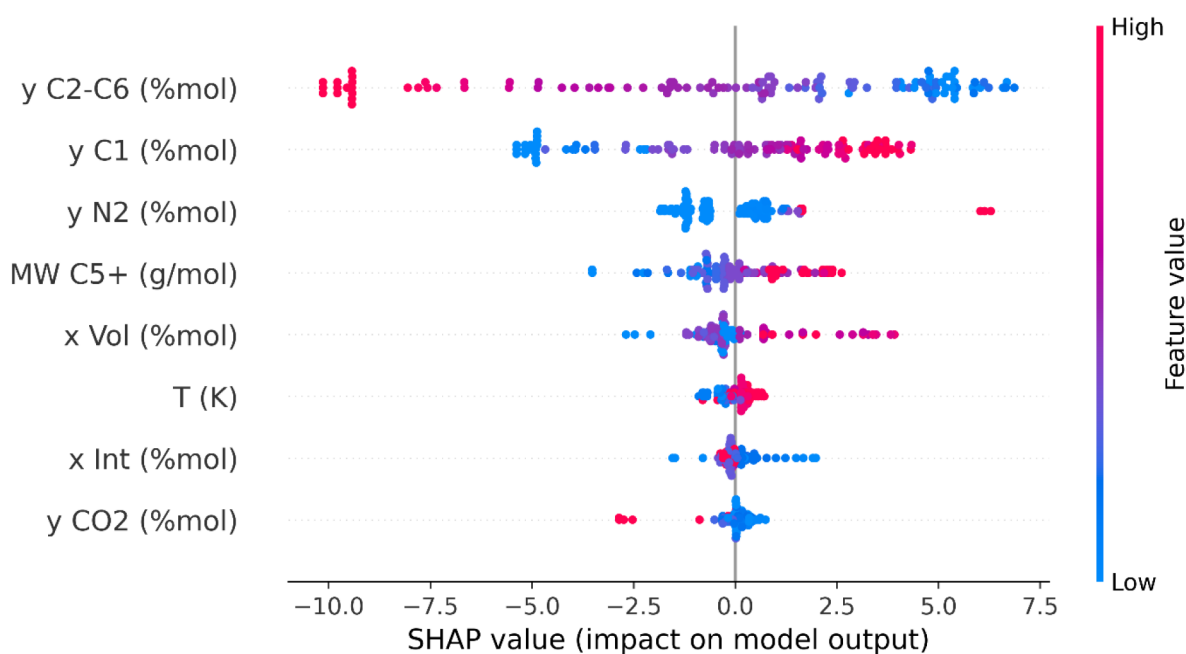


Figure 4—SHAP summary plot of the established ANN model

## Conclusions

This research presents a robust and data-driven framework for predicting MMP in hydrocarbon gas–oil systems using advanced machine learning techniques. By leveraging a carefully curated dataset of 118 experimental MMP measurements from a variety of published sources, we developed and validated two predictive models: an Artificial Neural Network (ANN) and the Extra Trees (ET) algorithm. The dataset included a broad range of gas compositions, with varying mole fractions of key hydrocarbon components such as C1 and C2–C6, ensuring that the models were trained on scenarios representative of real-world reservoir conditions. The ANN model demonstrated superior predictive accuracy compared to the ET model across all evaluation stages. It achieved a high coefficient of determination ( $R^2 = 0.9934$ ) and a low root mean square error (RMSE = 1.1883), indicating its strong capability to capture complex, nonlinear relationships inherent in gas–oil miscibility behavior. These findings were validated by visual cross-plots and residual analysis, which revealed tight alignment between predicted and experimental MMP values, as well as low and symmetrically distributed residuals. In addition to predictive accuracy, model interpretability was a key focus of this study. To open the black box of the ANN model and understand the influence of individual input features on the predicted MMP, we applied SHapley Additive exPlanations (SHAP), a powerful explainable

AI technique. The SHAP analysis not only confirmed known physical phenomena, such as the inverse impact of C2–C6 on MMP and the positive effect of increasing C1 content, but also provided a detailed, quantitative understanding of feature importance. These insights enhance the model's credibility and utility in practical field applications.

The results of this work show that coupling high-fidelity machine learning models with interpretability tools offers a practical, accurate, and transparent solution for MMP prediction. This approach enables reservoir engineers to optimize gas injection compositions and pressure strategies more effectively, potentially reducing costs and improving recovery efficiency in enhanced oil recovery operations.

Looking forward, the proposed methodology can be extended to more complex compositional systems or hybridized with uncertainty quantification and real-time data assimilation. Overall, this study lays a strong foundation for the integration of machine learning and explainable AI into the decision-making toolbox of petroleum engineers and EOR practitioners.

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## References

1. Chaudhary GZ, Ahmad I. Long-term dynamics of hydrocarbon fuels and economic growth in developing nations. *Journal of Energy and Environmental Policy Options* 2023;**6**:30–6.
2. Wang S, Li G, Fang C. Urbanization, economic growth, energy consumption, and CO<sub>2</sub> emissions: Empirical evidence from countries with different income levels. *Renewable and Sustainable Energy Reviews* 2018;**81**:2144–59. <https://doi.org/10.1016/j.rser.2017.06.025>.
3. Wei C, Jafari Raad SM, Leonenko Y, Hassanzadeh H. Correlations for prediction of hydrogen gas viscosity and density for production, transportation, storage, and utilization applications. *Int J Hydrogen Energy* 2023;**48**:34930–44. <https://doi.org/10.1016/j.ijhydene.2023.05.202>.
4. Nait Amar M, Djema H, Ourabah K, Alqahtani FM, Ghasemi M. Utilizing Artificial Intelligence Techniques for Modeling Minimum Miscibility Pressure in Carbon Capture and Utilization Processes: A Comprehensive Review and Applications. *Energy & Fuels* 2024;**38**:14891–924. <https://doi.org/10.1021/acs.energyfuels.4c02249>.
5. Belhaj H, Abukhalifeh H, Javid K. Miscible oil recovery utilizing N<sub>2</sub> and/or HC gases in CO<sub>2</sub> injection. *J Pet Sci Eng* 2013;**111**:144–52. <https://doi.org/10.1016/j.petrol.2013.08.030>.
6. Nait Amar M, Zeraibi N, Alqahtani FM, Djema H, Benamara C, Saifi R, et al Robust ensemble learning frameworks for predicting minimum miscibility pressure in pure nitrogen and gas mixtures containing nitrogen–crude oil systems: Insights from explainable artificial intelligence. *Can J Chem Eng* 2025. <https://doi.org/10.1002/cjce.25738>.
7. Ahmadi MA, Zendejboudi S, James LA. Developing a robust proxy model of CO<sub>2</sub> injection: Coupling Box–Behnken design and a connectionist method. *Fuel* 2018;**215**:904–14. <https://doi.org/10.1016/j.fuel.2017.11.030>.
8. Ding M, Yuan F, Wang Y, Xia X, Chen W, Liu D. Oil recovery from a CO<sub>2</sub> injection in heterogeneous reservoirs: The influence of permeability heterogeneity, CO<sub>2</sub>–oil miscibility and injection pattern. *J Nat Gas Sci Eng* 2017;**44**:140–9. <https://doi.org/10.1016/j.jngse.2017.04.015>.
9. Alatefi S, Agwu OE, Azim RA, Alkouh A, Dzulkarnain I. Development of multiple explicit data-driven models for accurate prediction of CO<sub>2</sub> minimum miscibility pressure. *Chemical Engineering Research and Design* 2024;**205**:672–94. <https://doi.org/10.1016/j.cherd.2024.04.033>.
10. Shafiei M, Kazemzadeh Y, Escrochi M, Cortés FB, Franco CA, Riazi M. A comprehensive review direct methods to overcome the limitations of gas injection during the EOR process. *Sci Rep* 2024;**14**:7468. <https://doi.org/10.1038/s41598-024-58217-1>.



11. Zhaomin L, Maen MH, Abdolhossein H-S. *Gas Injection Methods*. Elsevier; 2023. <https://doi.org/10.1016/C2019-0-04011-3>.
12. Jaubert J-N, Wolff L, Neau E, Avaullee L. A Very Simple Multiple Mixing Cell Calculation To Compute the Minimum Miscibility Pressure Whatever the Displacement Mechanism. *Ind Eng Chem Res* 1998;**37**:4854–9. <https://doi.org/10.1021/ie980348r>.
13. Fazlali A, Nikookar M, Agha-Aminiha A, Mohammadi AH. Prediction of minimum miscibility pressure in oil reservoirs using a modified SAFT equation of state. *Fuel* 2013;**108**:675–81. <https://doi.org/10.1016/j.fuel.2012.12.091>.
14. Benmekki EH, Mansoori GA. Minimum Miscibility Pressure Prediction With Equations of State. *SPE Reservoir Engineering* 1988;**3**:559–64. <https://doi.org/10.2118/15677-PA>.
15. Ghorbani M, Gandomkar A, Montazeri G, Honarvar B, Azdarpour A, Rezaee M. Experimental Investigation of Asphaltene Content Effect on Crude Oil/CO<sub>2</sub> Minimum Miscibility Pressure. *Periodica Polytechnica Chemical Engineering* 2020;**64**:479–90. <https://doi.org/10.3311/PPch.15980>.
16. Hamidpour S, Safaei A, Kazemzadeh Y, Hasan-Zadeh A, Khormali A. Calculation of IFT in porous media in the presence of different gas and normal alkanes using the modified EoS. *Sci Rep* 2023;**13**:8077. <https://doi.org/10.1038/s41598-023-35320-3>.
17. Nait Amar M, Zeraibi N, Djema H, Alqahtani FM, Benamara C, Saifi R, et al. A reliable model to predict mercury solubility in natural gas components: A robust machine learning framework and data assessment. *J Hazard Mater* 2025;**493**:138396. <https://doi.org/10.1016/j.jhazmat.2025.138396>.
18. Hemmati Sarapardeh A, Larestani A, Nait Amar M, Hajirezaie S. *Applications of artificial intelligence techniques in the petroleum industry*. Gulf Professional Publishing; 2020.
19. Youcefi MR, Wei W, Alqahtani FM, Djema H, Nait Amar M, Ghasemi M. Robust Ensemble Learning Models for Predicting Hydrogen Sulfide Solubility in Brine. *Energy & Fuels* 2024;**38**:21174–88. <https://doi.org/10.1021/acs.energyfuels.4c04354>.
20. Deffrenne P, Marle C, Pacsirszki J. The Determination Of Pressures Of Miscibility. Fall Meeting of the Society of Petroleum Engineers of AIME, SPE; 1961. <https://doi.org/10.2118/116-MS>.
21. Rathmell JJ, Stalkup FI, Hassinger RC. A Laboratory Investigation of Miscible Displacement by Carbon Dioxide. *All Days*, SPE; 1971. <https://doi.org/10.2118/3483-MS>.
22. Glasø Ø. Generalized Minimum Miscibility Pressure Correlation. *Society of Petroleum Engineers Journal* 1985;**25**:927–34. <https://doi.org/10.2118/12893-PA>.
23. Sebastian HM, Wenger RS, Renner TA. Correlation of Minimum Miscibility Pressure for Impure CO<sub>2</sub> Streams. *Journal of Petroleum Technology* 1985;**37**:2076–82. <https://doi.org/10.2118/12648-PA>.
24. Firoozabadi A, Aziz K. Analysis and Correlation of Nitrogen and Lean-Gas Miscibility Pressure. *SPE Reservoir Engineering* 1986;**1**:575–82. <https://doi.org/10.2118/13669-PA>.
25. Eakin BE, Mitch FJ. Measurement and Correlation of Miscibility Pressures of Reservoir Oils. *All Days*, SPE; 1988. <https://doi.org/10.2118/18065-MS>.
26. Jaubert J-N, Avaullee L, Souvay J-F. A crude oil data bank containing more than 5000 PVT and gas injection data. *J Pet Sci Eng* 2002;**34**:65–107. [https://doi.org/10.1016/S0920-4105\(02\)00153-5](https://doi.org/10.1016/S0920-4105(02)00153-5).
27. Al-Ajmi M, Alomair O, Elsharkawy A. Planning Miscibility Tests and Gas Injection Projects for Four Major Kuwaiti Reservoirs. *All Days*, SPE; 2009. <https://doi.org/10.2118/127537-MS>.
28. Adekunle O. *Experimental approach to investigate minimum miscibility pressures in the Bakken*. Colorado School of Mines, 2014.

29. Teklu TW, Alharthy N, Kazemi H, Yin X, Graves RM. Hydrocarbon and Non-hydrocarbon Gas Miscibility with Light Oil in Shale Reservoirs. SPE Improved Oil Recovery Symposium, *Society of Petroleum Engineers*; 2014. <https://doi.org/10.2118/169123-MS>.
30. Hawthorne SB, Miller DJ, Jin L, Gorecki CD. Rapid and Simple Capillary-Rise/Vanishing Interfacial Tension Method To Determine Crude Oil Minimum Miscibility Pressure: Pure and Mixed CO<sub>2</sub>, Methane, and Ethane. *Energy & Fuels* 2016;**30**:6365–72. <https://doi.org/10.1021/acs.energyfuels.6b01151>.